Subtractive Mixture Models via Squaring: Representation and Learning

Paper Code

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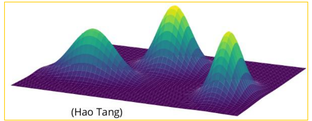
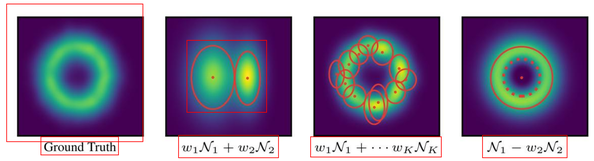
0 Mixture models

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \quad \text{subject to} \quad w_i \geq 0 \quad \sum_{i=1}^{K} w_i = 1$$



✗ components can only be added together!



Fewer components with subtractions

Questions? **...Contributions!** 1 2 3

1 How to learn subtractive mixture models?

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \qquad w_i \in \mathbb{R}$$

How to ensure $p(\mathbf{X})$ is non-negative?

$\Longrightarrow$ Impose ad-hoc constraints over the parameters

✗ challenging to derive in closed-form [1][2][3]

2 How much more expressive are they?

with respect to traditional additive-only mixtures

3 What is their relationship with other models?

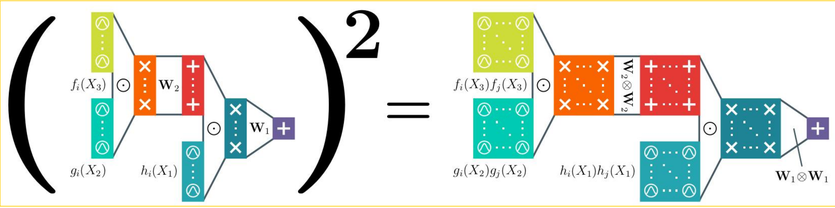
understanding why they are expressive ...

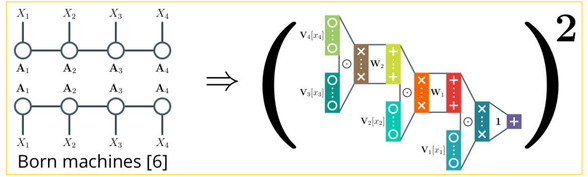
... and why they support tractable inference

TL; DR

"We learn exponentially more expressive mixture models with subtractions, by squaring deep tensorized mixtures"

Learning deep subtractive mixtures by squaring layers of a deep circuit





$$p(\mathbf{x}) \propto \kappa(\mathbf{x})^\top \mathbf{A}\, \kappa(\mathbf{x})$$

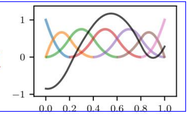
PSD models [7] $\Rightarrow \sum \left(\mathbf{w} \right)^2$

1 Squaring mixtures ...

$$p(\mathbf{X}) \propto \left(\sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \right)^2 = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \, p_i(\mathbf{X}) p_j(\mathbf{X})$$

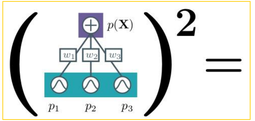
Renormalization:

$$Z = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \int p_i(\mathbf{X}) p_j(\mathbf{X}) \mathrm{d}\mathbf{X}$$



Tractable marginalization is supported by exponential families [2] and splines components

Build deep mixtures with layers as "Lego blocks"

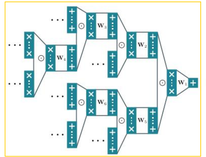


... by squaring circuits

2 Theorem. exponential separation [4] [5]

There is a class of distributions $\mathcal{F}$ over variables $\mathbf{X}$ that can be compactly represented as a shallow squared mixture with negative weights, but the smallest structured decomposable additive-only mixture of any depth computing any $F \in \mathcal{F}$ has size $2^{\Omega(|\mathbf{X}|)}$.

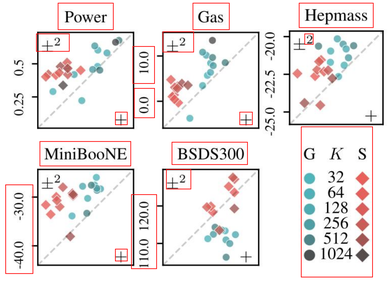
Deep additive-only mixtures



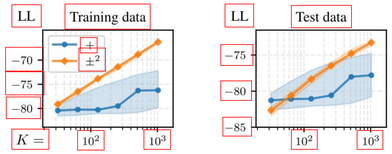
Squared subtractive mixture model

3 Understanding the expressiveness of other models in a unifying framework

Density estimation



GPT2 distillation [8]



References

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